

EM Seismographs

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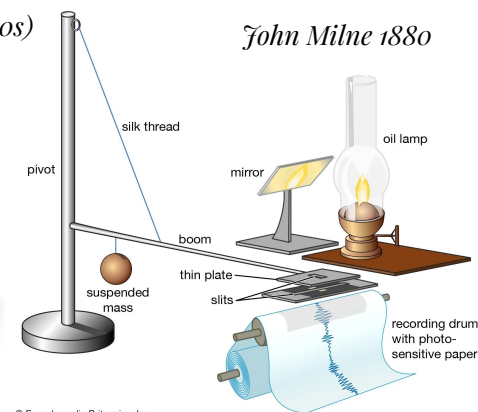
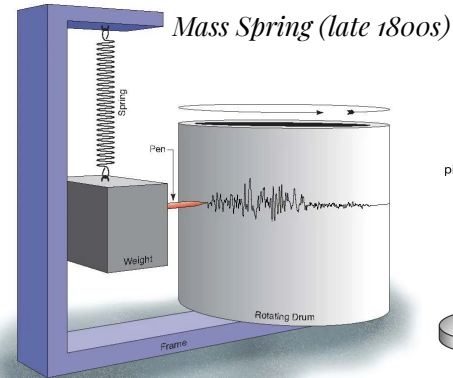
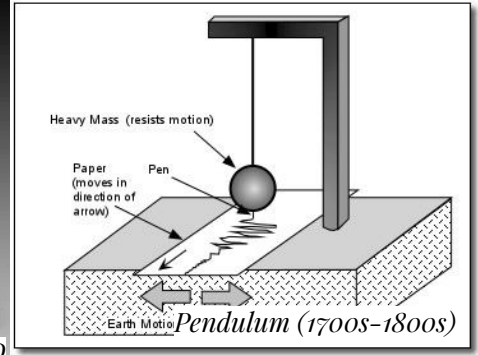
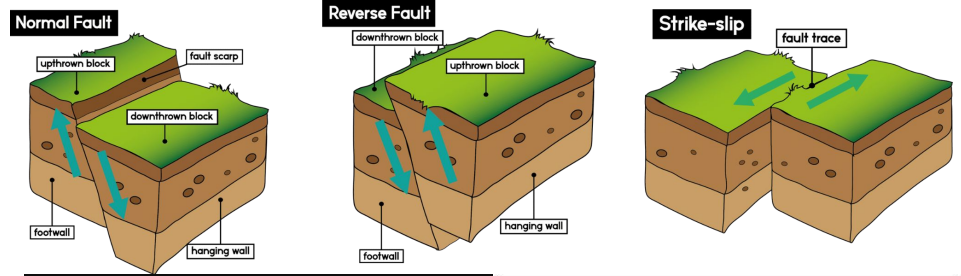
Seismograph Principles

Idea: Want to detect vibrations (seismic waves) to create a seismogram

Seismograph Use Cases:

- Find epicenter
- Measure magnitude and intensity
- **Early warning** systems
 - P-Waves quicker than S-Waves
- Studying Earth's interior

Ancient idea with **many** iterations!



Physics of Electromagnetic Seismographs

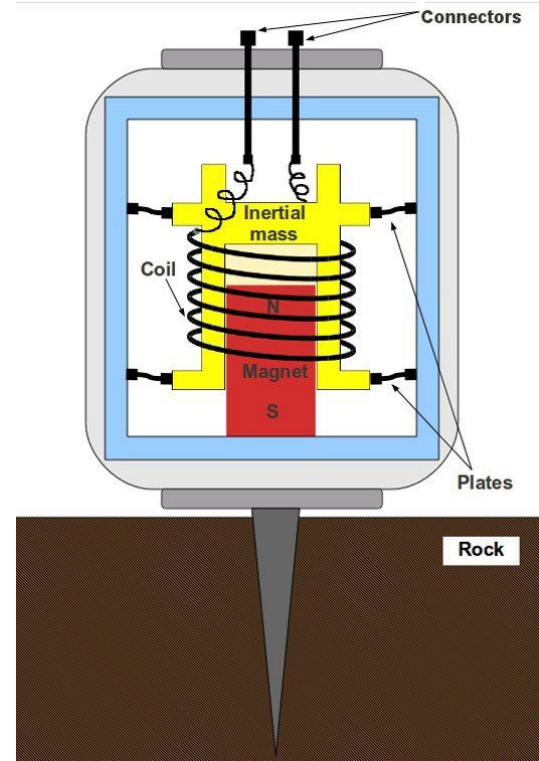
Creates a voltage by using either a fixed (to the ground) **magnet** and floating **coil** or vice versa

When the ground vibrates and a component moves relatively, a **voltage** (by **Faraday's Law**) is then measured and correlated with the movement of the ground

Machines are calibrated to produce velocity data

$$m \frac{d^2 x}{dt^2} + kx = 0$$

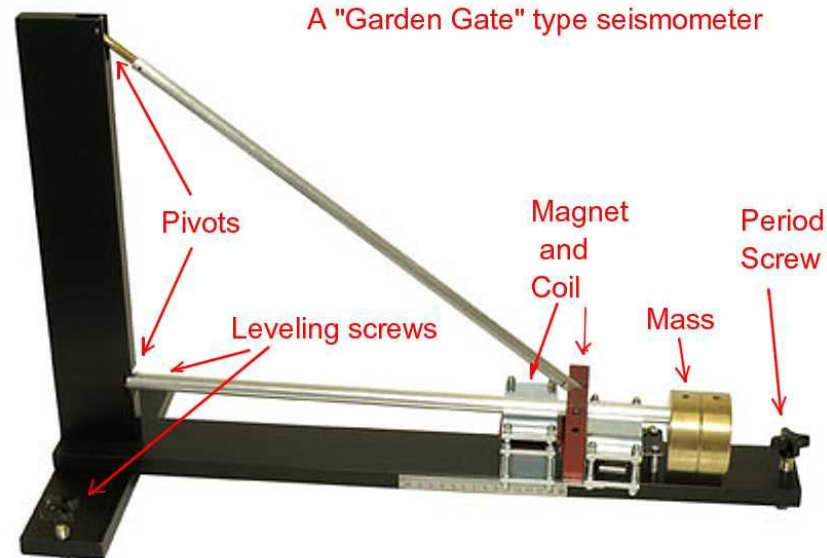
$$\varepsilon = \frac{d\varphi_B}{dt}$$



PHYSICAL MODEL

Real Calculation

Seismometers are individually measured for their “**seismometer constant**”, and are built to have **linear** responses to ground movement. The seismometer constant (**S**) is in $V/(m/s^2)$. Using the equation $v = S/V$, the **velocity** of the ground can be determined



Modern Seismometers

Force-Balance Seismometers:

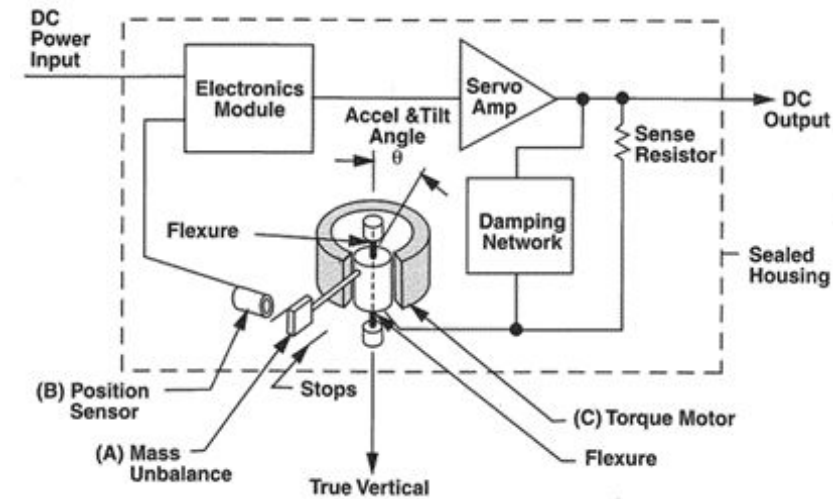
Measures force required to keep a mass **motionless** ($F=0$) relative to the frame, yielding ground acceleration

Fiber-Optic / Laser Seismometers:

Fiber-optic cables change in **length** during an earthquake, causing light to either:

- Travel longer distances
- Travel shorter distances
- Be scattered differently

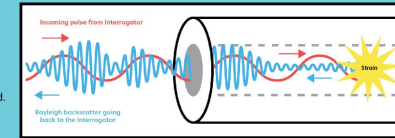
This can be measured!



Distributed Acoustic Sensing

DAS can detect many things, including drilling activity, ships, or marine life in the ocean, as well as seismic activity, landslides, or even movements of glacial ice.

Phase shifts between outgoing and incoming light waves can tell us the location in the cable where the strain occurred.



Interrogator

Volcanic activity

Earthquake

The interrogator records the seismic waves, converts it into raw data, and locates the sound sources.

Rayleigh backscattering occurs at imperfections or variation in the fiber. The light pulse is disrupted, and some will go back to the interrogator for processing and analysis.

Earthquake

The fiber optic cable provides a spatially continuous measurement sensing any disturbance along the length it covers, which can be tens of kilometers long.

Conclusion

EM Seismometers were a **large** development in the world of detecting seismic activity, and are used throughout the world to monitor earthquakes, possible tsunamis, drilling, and even nuclear bomb tests

Their development reflects an **ongoing** process of engineering innovation, with each generation of seismographs becoming more **accurate**, **sensitive**, and **reliable** than the last, demonstrating how continuous improvements in technology have expanded our ability to understand and monitor Earth's activity

Thank you!

